Experiment 6 Day 3: Conversion of Borneol to Camphor

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Introduction

In this experiment, borneol and glacial acetic acid were combined in a long-neck, round-bottom flask before being refluxed. Once these reagents had dissolved, an aqueous sodium hypochlorite solution was added dropwise over the next 35 minutes. After this reflux was completed, the product was allowed to cool before being extracted with methylene chloride; the product was then washed sequentially with saturated sodium bicarbonate and water. The organic solution was then dried over anhydrous sodium sulfate before being evaporated over a water bath. The finished product was then weighed and the percent yield was determined.

Materials and Methods

First, the sand bath was plugged into the 10 Amp outlet and was set to 60. Next, in a 10 mL long-neck round-bottoms flask 0.3 g of borneol product and 0.3 mL of glacial acetic acid were added. A reflux was assembled and the water bath range was kept between 40-50℃. Once these reagents had dissolved, 6 mL of aqueous sodium hypochlorite solution (bleach) was added dropwise through the condenser over the next 35 minutes. After each addition, the solution was given a gentle swirl to ensure the chemicals mixed thoroughly. After all of the bleach was added, the solutions was refluxed for an additional 15 minutes. The reaction was then allowed to cool to room temperature before being transferred to a centrifuge tube. The camphor was then extracted utilizing two portions (2 mL each) of methylene chloride. The combined methylene chloride layers were then neutralized with about 2 mL of saturated sodium bicarbonate. Next, the organic solution was washed with 2 mL of 5% sodium bisulfite followed by another wash with 2 mL of water. The organic solution was then dried over anhydrous sodium sulfate. The methylene chloride was then evaporated off of the product over a water bath not exceeding 50℃. The collected crystals were then weighed and a percent yield was determined.

Results

Theoretical Yield (g)	~0.3
Experimental Yield (g)	0.143
Percent Yield (%)	47.7

Table 1. Theoretical Yield, Experimental Yield, and Percent Yield

Discussion

A quantitative assessment of this experiment showed that somewhere throughout the experiment a mistake was made. While there was a good yield of product from this experiment, it was much lower than the theoretical yield value. Possible sources of error in this experiment include inadequate refluxing or overrefluxing during step 3 of this experiment, inaccurate measurements of any of the liquid reagents or the solid borneol, inadequate extracting of the product (utilizing methylene chloride) or washing of the product (utilizing 5% sodium bisulfite and water), or inadequate drying of the product utilizing anhydrous sodium sulfate. Of the sources of error mentioned, the one to have most likely occurred was the overreluxing of the product during step 4. This was the most likely source of error as overrefluxing of the reaction may have caused a loss in the total product formed.

References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Lab Manual [lab manual]. Auburn (Al): Auburn University.